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ORIGINAL ARTICLE

Emotional evaluation for pictures displayed with small FOV telescope environment in virtual reality headset

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Abstract
We carried out an experiment to examine emotional effect associated with images displayed in small field of view (FOV) telescope like virtual screen environment with VR-HMDs. We manipulate FOV of a virtual camera in VR to change the view of contents. Decreasing FOV makes view of angle narrow, and therefore, the view in VR looks like telescope, in which environment the view greatly moves with even small HMD movement. With smaller value of FOV, the view in VR is more sensitive to HMD movement and requires subjects to keep concentrating not to move their heads to watch target contents properly. Stronger emotional responses were observed with smaller value of FOV in the experiment. The result indicates the possibility of controlling effectiveness of the same contents with the same VR-HMD. This result is expected to improve a wide range of medical usage of VR contents.

Keywords VR · Emotion · Psychology · FOV · Self-assessment manikin · IAPS

1 Introduction

Low-cost and high-quality virtual reality head mounted displays (VR-HMDs) are getting popular and used for various applications in these days. As more people use VR-HMDs for longer time, psychological impacts are considered as one of the important factors for healthy growth of VR industry.

Side effects of VR such as nausea, oculomotor, and disorientation symptoms are one of the concerns for long period of usage of VR. Sharples et al. show induced more symptoms and effects comparing to the other displaying devices such as desktop, projection, and theatre screen [1].

On the other side, positive psychological effect of VR can also be used for medical usage. Anxiety and pain management using VR are widely studied and thought that VR could be part of the future of clinical psychology. Impact of psychological factors on the effectiveness of VR-based analgesia to distract from pain was analyzed and concluded the presence, fun, and anxiety influence VR efficacy [2]. Hunter G et al. applied VR as analgesic for burn injured patients to reduce procedural pain. Interactive immersive VR contents were used and reported 35–50% reductions in procedural pain [3]. The logic of pain reduction was discussed that humans have limited attentional capacity and patients have less attention available to process incoming signals from pain receptors while in VR [4].

BiPSEE Inc., and Medical Corporation Iryouhoujin Koukeikai started using VR-HMD with kid's animation for children in dental clinics, while treatment of tooth and getting positive feedback about the usefulness of keeping the child patients calm especially for children with attention-deficit hyperactivity disorder (ADHD). Gongsook shows the significant role of virtual reality in improving ADHD and reducing behavioral symptoms and problems [5].

The previous studies show that VR affects our emotions and body, and it works as an efficient anxiety and pain management tool to patients. However, the problem is that the efficacy of the result depends on hardware and contents. Low latency virtual environment affects physiological response, simulator sickness, and self-reported sense of presence [2]; however, the VR-HMD hardware improvement is needed to lower latency of VR-HMD. Contents

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related to snow works significantly for burn injured patient as a pain reduction tool [6], and kids animation can direct children's attention in VR. However, the same contents may not work well for different category of people.

To explore a way of effective emotional responses which does not depend on hardware nor contents, we carried out two types of experiments with FOV manipulated virtual screens in VR-HMD. Small FOV virtual screen was prepared and named telescope screen for the experiments. The results of the first experiment show that watching pictures with the telescope screen affects greater emotional responses. The result of the second experiment indicates that more attention of subjects was directed to the screen, which supports the result of the first experiment.

Unlike FOV of HMD hardware, FOV of virtual camera can be manipulated by software regardless of its displaying HMD hardware. This novel method to enhance emotional responses and to direct attentions of people is expected to use in anxiety and pain management as well as e-learning, rehabilitation of ADHD, medical preparation, etc., because the method does not depend on hardware nor contents.

2 Emotional evaluation experiment

We carried out the first experiment to examine emotional effect with pictures displayed in different FOV virtual screen environment in VR-HMDs and evaluate the emotional response with normal and small FOV telescope screen using International Affective Picture System (IAPS) [7] and Self-Assessment Manikin (SAM) [8] method.

Kawai et al. show the enhancement of emotional arousal by stereoscopic images with IAPS and SAM in a previous study [9].

2.1 Methods

2.1.1 Participants

Twelve volunteers (nine males and three females) were participated in the experiment as subjects. Ages are 21–61 years.

Table 1 Parameters of two virtual screen environments

Type	Screen size (m)	FOV	Distance (m)
Normal	8×6	108°	4
Telescope	8×6	2°	1200

2.1.2 Setup

2.1.2.1 Virtual screen and FOV Field of view (FOV) of a virtual camera was manipulated in VR-HMD to change the view of contents for subjects. Decreasing FOV makes view of angle narrow; therefore, the view in VR is similar to telescope, in which environment the view moves a lot with even small movement.

With smaller value of FOV for a virtual camera, the view in VR is more sensitive to HMD movement and requires subjects to keep concentrating not to move their heads to watch target images properly. Two types of virtual screen environments were prepared for the experiment (Table 1).

Both screens are 8×6 m size square flat screen in virtual environment and displayed as approximately same size and fitted to HMD screen width. However, normal screen was located at 4 m from eyes in virtual environment, and the FOV for a virtual camera was set to 108°, so that the virtual environment was similar to an environment in which actual 8×6 m size screen was located at 4 m from subjects. On the other hand, telescope screen was located at 1200 m from eyes in virtual environment with 2°FOV virtual camera. The screen environment works like telescope view with the screen located at 1200 m distance from subjects (Fig. 1).

The created visions with both normal and telescope screens look identical when subjects face the screen, but the screen moves more in telescope environment with HMD movements (Fig. 2). In the telescope screen environment,

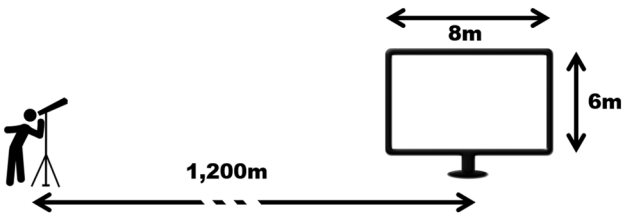


Fig. 1 Schematic drawing of the telescope virtual screen environment

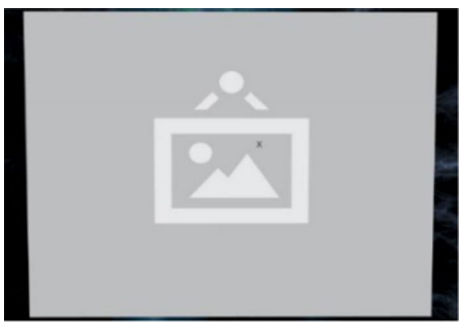


Fig. 2 Created vision in HMD. Pictures were displayed in the experiment

only 2° movement of the HMD makes the screen out of the view.

2.1.2.2 HMD Smartphone-based HMD was built up with 5.8-inch smartphone and Cardboard type VR component kit. The HMD is 18.5 × 16.1 × 1.1 cm size and 224 g light weight including the smartphone. Subjects hold the HMD in front of their face, and the HMD displays contents based on its gyroscope. The HMD is connected to a PC, which controls the contents of the HMD and records logs of the experiment. We used monocular lens goggles instead of binocular lens used for typical VR goggles, since binocular lens VR-HMD is not recommended for people under 13 years old because of the risk of crossed eyes. This study is expected to be used for children as well as adult people.

2.1.2.3 SAM Self-assessment manikin (SAM) was developed by Lang and used to evaluate the emotional response of the subjects. SAM is a widely used non-verbal assessment technique which measures pleasure, arousal, and dominance with cartoon characters graphic scales. Manikin images for rating (Fig. 3) were prepared to measure arousal level in nine scales and selected by subjects after a picture appears in HMD.

Morris shows that SAM is effective as a cross-cultural measurement of emotional response [10]. The previous studies show that the relationship between affective pictures and arousal. Gil et al. show that emotional time distortions are induced by arousal [11].

2.1.2.4 Pictures International Affective Picture System (IAPS) was used as emotional stimuli in the experiment. IAPS is a widely used database of pictures designed to provide a standardized set of pictures for studying emotion and attention in psychological research. Lang et al. show its effectiveness as an affective rating tool.

Pictures of the experiments were selected from IAPS photo data set. All pictures of IAPS are rated in the reactions of arousal, valence, and dominance. Pictures were sorted by arousal rating and selected 20 most high rated and 20 most low rated pictures. The two groups of pictures are named "Intense" and "Normal" group. Pictures of Intense group were expected to stimulate subject's emotion, and pictures of Normal group were expected to keep subject's emotion calm.

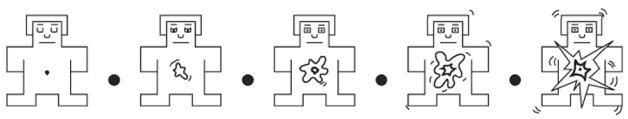


Fig. 3 Self-assessment manikin (SAM) scales for arousal

2.1.2.5 Software Software for the experiment was installed to a smartphone, and the device was attached to a goggle, which was used as a VR-HMD. During the experiment, randomly selected images from a list are displayed after a countdown on a virtual screen in the VR-HMD. FOV of a virtual camera in VR also randomly changes, and relationship between FOV and emotional response was recorded.

Software randomly selects a type of the virtual screen and a picture to display. Two types of the virtual screen "Normal screen" and "Telescope screen" were selected by 50% probability, and the two photo set groups "Intense" and "Normal" were also selected by 50% probability. The software chooses a picture to display at random from the selected group of the photo set. Selected type of the virtual display was used to show the selected picture in the HMD by the software.

2.1.3 Procedure

The experiment was carried out by two people in a calm room, and lights were tuned off, so that subjects can be relaxed during the experiment. Subjects were asked to wear acoustic earmuff to reduce environmental noise, while they use VR-HMD (Fig. 4). One person controls PC and another person recorded the result. Beginning of the experiment, overview of the procedure was explained to subject about what they need to do, but the expected outcome or the purpose of the experiment was not explained to the subjects. Subjects were mentioned that unpleasant pictures may appear on screen and they may stop the experiments if they feel uncomfortable. Subjects were also asked not to tell other people about the detail about the experiment because of prior information control of the psychological experiment.

One session was composed with 20 picture displays. In a session, a loading image appears for 8–13 s and a picture appears for 6 s. The duration of the loading image varies from 8 to 13 s, because subjects should not know exactly



Fig. 4 The condition of the experiment

when the picture appears. After the picture disappears, the subject indicates their arousal level with the SAM picture.

Picture type, screen type, and arousal level were recorded along with picture and subject number.

2.2 Results and consideration

Emotional responses were recorded in 227 trials for 12 subjects. Two subjects stopped their sessions at their 11th and 18th trials due to a feeling of unwellness. The data before they stopped their session were included in the result. The interruption is considered to be due to the emotional response of the experiment rather than VR sickness, since the two subjects use VR-HMD for their works and tolerant of it. The screen type and picture type were determined at random by the software at each trial. Actual numbers of observation were not distributed equally. Normal pictures appeared 61 times on normal screen and 45 times on telescope screen. Intense pictures appeared 59 times on normal screen and 62 times on telescope screen.

t test result with all pictures' data set indicates that watching pictures in telescope screen environment makes subject feel more arousal.

To understand more about the relationship of the virtual screen environment and the arousal level, recorded data were also analyzed with the categorized pictures set. Analyzed data with normal and intense pictures' data set show that type of virtual screen environment does not affect emotional responses significantly with normal pictures.

Injured pictures data set was selected from intense pictures for further analysis, which include mutilation and tumor pictures. The mean of the arousal ratings of injured pictures set was higher than the ratings of intense pictures set.

F test was performed first to determine the equality of variances of the data. Welch *t* test was performed assuming unequal variances for all pictures' data set. On the other hand, *t* test assuming equal variances was performed for Normal, Intense, and Injured pictures data set.

The results of *t* test were $p = 0.04$ for injured pictures set, $p = 0.05$ for intense pictures set, and $p = 0.46$ for normal pictures set, which indicate that telescope screen environment makes subjects feel more arousal with higher arousal rating pictures (Table 2).

VR-HMD was used for the experiment; however, the contents used in the experiment were ordinal two-dimensional pictures. Emotional evaluation with 360 photos or immersive three-dimensional contents needs to be examined in the further study to apply this method to the recent VR contents.

To improve the efficiency of the method, relationship between the value of FOV and the arousal level also needs to be examined in the further study (Fig. 5).

3 HMD stability evaluation experiment

The second experiment was carried out to explain the reason why subjects had stronger emotional response with telescope screen. Stability of HMDs which were evaluated with normal and telescope screen to see more concentration is needed with telescope screen to watch contents.

3.1 Method

3.1.1 Participants

Twelve volunteers (five males and seven females) were participated in the experiment as subjects. Ages are 15–52 years.

3.1.2 Setup

3.1.2.1 HMD Instead of handheld Cardboard type setup used in the emotional evaluation experiment, wearable HMD was built up with 5.8-inch smartphone and VR Headset (ELECOM P-VR1G01WH). The HMD is $20.3 \times 19.3 \times 11.8$ cm size and 482 g light weight including the smartphone. Video contents were used instead of pictures in this experiment. Audio of video contents was played from a connected PC.

3.1.2.2 Video In the previous experiment, subjects often moved HMDs when the pictures appeared by surprising.

Table 2 *t* test result and statistics of recorded arousal level for selected picture data set

Used pictures' data set	Normal screen (FOV:108°)			Telescope screen (FOV:2°)			*F* test result	*t* test result
	n	*M*	SD	*n*	*M*	SD		
All pictures	120	2.4	2.0	107	3.0	2.6	$F(119,106) = 0.73, p < 0.05$	$t(196) = 1.65, p < 0.05$
Normal pictures	61	1.6	0.9	45	1.6	1.0	$F(60,44) = 0.63, p > 0.05$	$t(104) = 1.66, p > 0.05$
Intense pictures	59	3.3	2.3	62	4.0	2.9	$F(61,58) = 0.65, p > 0.05$	$t(119) = 1.66, p = 0.05$
Injured pictures	22	3.7	2.8	17	5.5	3.2	$F(21,16) = 0.46, p > 0.05$	$t(37) = 1.69, p < 0.05$

n = Number of saqmples, *M* = mean, *SD* = standard deviation

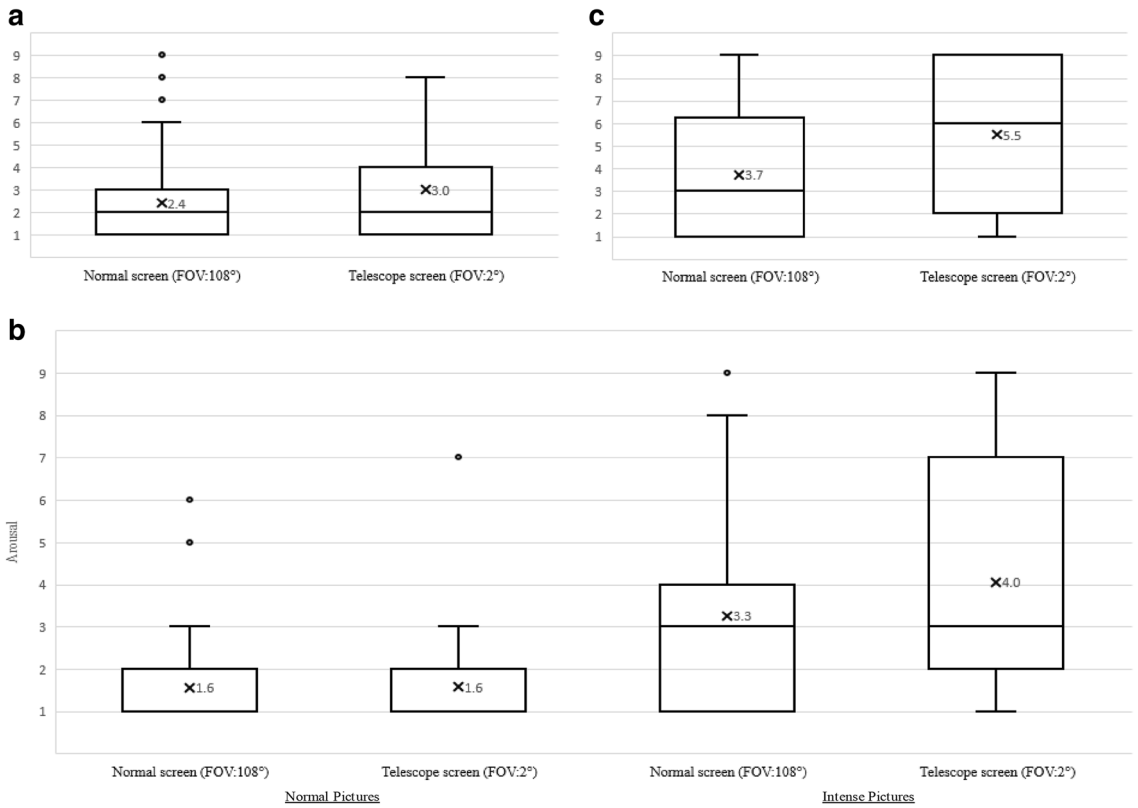


Fig. 5 **a** Arousal level with different FOV virtual screens for all pictures. **b** Arousal level with different FOV virtual screens for normal and intense pictures. **c** Arousal level with different FOV virtual screens for injured pictures in intense picture group

To test the HMD stability just caused by the screen type, video contents were prepared for this experiment instead of IAPS pictures. One video content was selected and extracted to 100 s. The same video data were used for all the subjects with both normal and telescope screen in the experiment.

3.1.2.3 Software Video playback and screen types are controlled by designated software to evaluate how much subjects keep HMD stable. In a session of the experiment, physical stability of HMD was recorded by the software. The software records psychical rotation angle of HMD on vertical and horizontal axis in 10 Hz using gyroscope sensor in the smartphone-based HMD. When subjects tilt the HMD by certain degrees, the angle of the rotation from its initial position is stored in a log file. The rotation angle around vertical axis is recorded as X, and the angle around horizontal axis is recorded as Y. The software switches the type of screen randomly every 10 s and resets the screen position to its initial location against HMD. In the 100 s session, the software changes the screen ten times at most and record 1000 points of data for stability analysis.

3.1.3 Procedure

The experiment was carried out in the same environment as the previous experiment. Only difference is that subject did not wear acoustic earmuff at the experimental session, because subjects watch a video with audio.

Subjects were told to watch a video with the HMD for 100 s, but no further information was told. Subjects were asked not to tell the other people about the detail about the experiment after the session.

Stability information and the screen type were recorded along with the subject information in a log file. Logged data were analyzed after the experiment.

3.2 Results and consideration

Figure 6 is a scatterplot of all the recorded data for 12 subjects consists of 12,000 data point. The plot indicates that HMDs are more moved when they watch video with normal screen comparing to telescope screen.

Distance from the center of the plots was averaged for each screen type as an indicator of stability, and the result

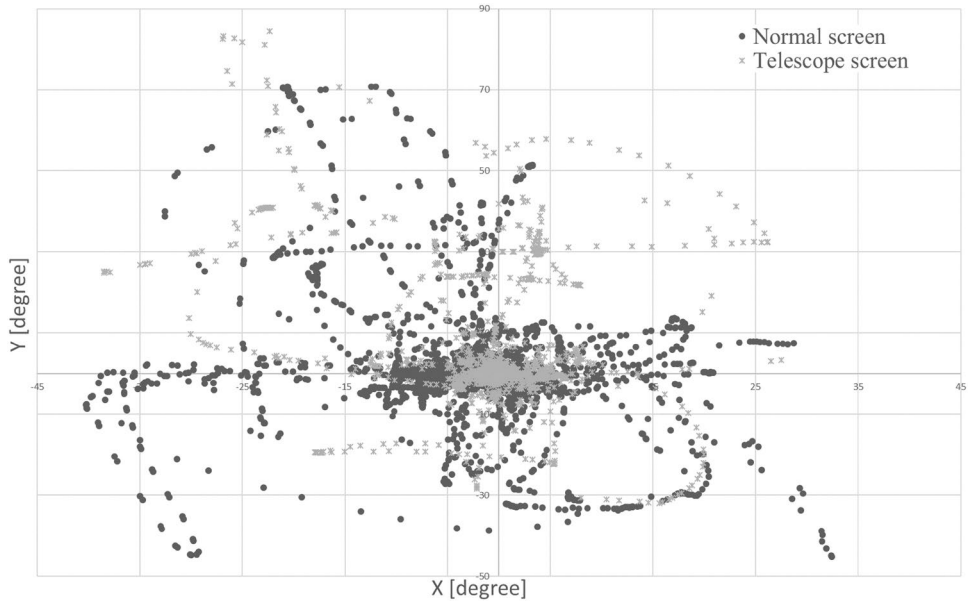


Fig. 6 Scatterplot of HMD angles of all the recorded data for 12 subjects. X as rotation angle around vertical axis and Y as rotation angle around horizontal axis

Table 3 Average of distance from center

Type	Stability of HMD
Normal	3.0718
Telescope	1.5389

Smaller value is more stable

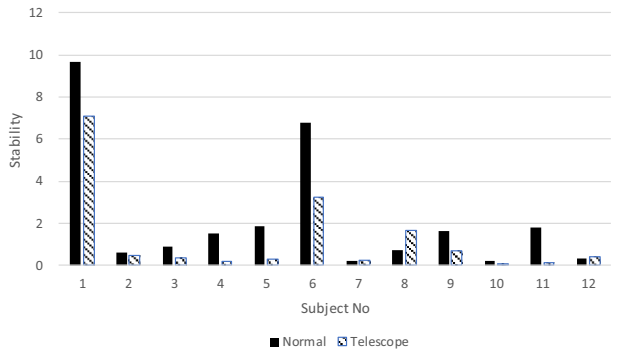


Fig. 7 Plot of stability data of both normal and telescope screen for each subject

Table 4 Stability of HMD and *t* test result

Subject	Normal	Telescope	Difference
1	9.6500	7.0990	2.5510
2	0.5972	0.4911	0.1061
3	0.9068	0.3688	0.5380
4	1.5309	0.1861	1.3448
5	1.8531	0.3136	1.5395
6	6.7575	3.2583	3.4992
7	0.2103	0.2351	−0.0248
8	0.7495	1.6749	−0.9254
9	1.6139	0.6717	0.9422
10	0.2162	0.0988	0.1174
11	1.7822	0.1395	1.6427
12	0.3131	0.4075	−0.0943
Mean	2.1817	1.2454	
Variance	8.6504	4.2198	
df		11	
t stat		2.5958	
$p(T \leq t)$ one-tail		0.0124	$p < 0.05$

shows that the value of normal screen is more than doubles of the value of telescope screen (Table 3).

Stability of HMD was calculated with the following formula where x_n and y_n indicate nth data of recorded data:

$$\text{Stability} = \frac{1}{n}.$$

Stability data of both normal and telescope screen of each subject were plotted in Fig. 7. Normal screen always appeared on the display of the HMD during the experiment; however, telescope screen was out of range for 20.9% of the total time. When subjects moved HMDs more than 2°, the screen went out from subjects' field of vision, and they needed to move their HMDs back to the center to keep the screen visible.

To analyze the data statistically, average of the distance was calculated for each subject. The data were evaluated with paired *t* test, and the result is significant (Table 4).

Even though the statistical analysis shows the significant difference of the two data sets, three subjects 7, 8, and 12 had less motion with normal screen, which is not our expectation. However, the motions with both normal and

telescope of subject 7 and 12 were small comparing to the other subjects, so the data seem not affected to the result of *t* test. To investigate the reason of the unexpected data for subject 8, HMD movement data are plotted as Fig. 8. One big HMD movement was observed when the screen type was telescope. We assume that this one exceptional movement caused unexpected result of subject 8.

The overall result shows that subjects keep VR HDMs more stable when video contents appeared on the telescope screen. Keeping VR-HMD in the same position requires much concentration and directs much attention to the screen. Therefore, the telescope screen made subjects focus on the screen and induced greater emotional responses, which can explain the result of the first experiment.

4 Conclusions

The results with *t* test statistically show that emotional responses were enhanced with only displaying software with manipulated FOV in the same VR-HMDs. The result indicates that the method directs attentions to screen and induce concentrations, as well.

This hardware and contents independent novel method could apply lots of VR applications including psychological therapy, learning, rehabilitation of development disorders, etc., without expensive hardware or contents' cost.

Extremely small number of FOV was used in this study to show the concept of the method and its result. The relationship of the value of FOV and its effectiveness needs to be examined in the further study.

We continue to study the method with small FOV telescope screen in VR-HMD to enhance emotional responses also with intermediate value of FOV as well as various types of contents, and applications using the method.

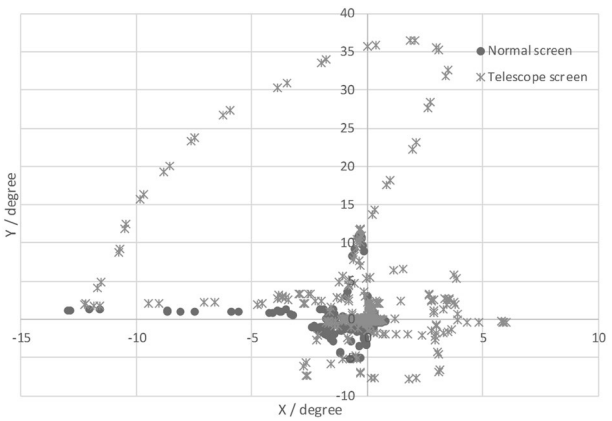


Fig. 8 Scatterplot of HMD movement of subject 8

Appendix

IAPS picture numbers used for the emotional evaluation experiment.

Normal pictures: 2190, 2840, 3170, 5130, 5740, 5800, 7000, 7004, 7006, 7010, 7020, 7031, 7060, 7080, 7110, 7187, 7217, 7490, 7491, and 7950.

Intense pictures: 3000, 3010, 3060, 3069, 3080, 3170, 4220, 4290, 4668, 4800, 5621, 6230, 6350, 6550, 8030, 8179, 8185, 8492, 9410, and 9940.

Injured pictures: 3000, 3010, 3060, 3069, 3080, and 3170.

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